

Vol 3 Chapter 11 — Nuclear Decay: g Substrate Field Exponent

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2026-05-23 Saturday

Volume 3, Chapter 11 — head-note (2026-09-19)

Head-note added 2026-09-19 (Keeper, own author pass corrected on Grace R154 N2 / K1913). Body unedited except the bracket in the SP-29-1 paragraph: the Cs-137 cavity effect is registered at 0.199 % (1990 ppm), three orders above the “parts-per-million” this chapter said; the smaller phrase had no source. Status: live, not tested (register N2; class bounds: DDEP half-life 30.024(11) y; Gran Sasso no modulation $> 9.6 \times 10^{-5}$ — see Grace’s pin note 2026-09-19).

Chapter 11 — Nuclear Decay

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Nuclear decay — alpha decay, beta decay, gamma decay, neutron emission — is the substrate’s weak-sector response to unstable nuclear configurations. Each decay channel has a characteristic substrate-mechanism reading via the BST primary $g = 7$ gauge-dimension integer.

11.1 Beta decay and the g-exponent

Beta-decay rates scale as $\sim G_F^2 E^5$ in standard physics, with G_F the Fermi coupling constant. The substrate-mechanism reading: the E^5 phase-space factor inherits the $n_C = 5$ substrate-dimension primary, and G_F is substrate-derived from the weak-sector gauge-bundle structure (Volume 2 Chapter 2).

Per-nuclide beta-decay rates match experimental measurements at percent precision across the catalog.

11.2 Alpha decay and the Gamow factor

Alpha decay rates involve the Gamow tunneling factor — exponentially suppressed by the nuclear-Coulomb barrier. The substrate-mechanism reading: the Gamow factor’s structure inherits from the substrate’s Casimir-vacuum coupling at the nuclear scale.

11.3 Gamma decay multipoles

Gamma decay multipole rates follow the $\alpha^{\text{BST primary}}$ exponent pattern of Chapters 8-9, with the multipole order L producing α^{2L-1} suppression.

11.4 The Cs-137 SP-29 falsifier

The SP-29-1 H4 Cs-137 paper-grade proposal (Elie + Grace, Casey directive May 18, 2026) targets precision measurement of Cs-137 beta-decay rates as a substrate falsifier. The substrate prediction: the standard exponential-decay model receives a substrate-vacuum correction at the parts-per-million level [*Corrected with this head, 2026-*

09-19 (K1913 on Grace R154 N2): the registered number is $3/1507 = 0.199\% = 1990\text{ ppm}$ (T2362; SP-29-1 proposal v0.3), not “parts-per-million”; the phrase had no source. The effect is *LIVE*, NOT TESTED — falsifier register v0.18 row N2.], testable with sub-MeV-precision gamma-ray spectroscopy of the Cs-137 daughter.

11.5 What comes next

Chapter 12 — Bridge Synthesis — cross-links Vol 3’s nuclear/atomic content to Vol 4 (cosmology, BBN) and Vol 9 (condensed matter).

Where to look this up: SP-29-1 Cs-137 proposal: `notes/SP29-1_H4_Cs137_paper_grade_proposal.md`. For standard nuclear decay: Krane Chapters 8–10.